# PEF 6000 - Special topics of dynamics of structures
# **Module 1 - Basics on dynamics of structures**

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# Outline

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# Objectives and references

- To introduce to basic aspects related to analytical mechanics and dynamics of structures;
- The topics addressed in the class are discussed in greater depth in other graduate courses (PEF 5916 and PME 5010);
- Examples of references:
  1. Clough, R.W. & Penzien, J., 1975. *Dynamics of Structures*. McGraw Hill.
  2. Lanczos, C., 1986. *The variational principles of mechanics*. Dover publications.
  3. Mazzilli, C.E.N., André, J.C., Bucalem, M.L. & Cifú, S., 2016. *Lições em mecânica das estruturas: Dinâmica*. Edgard Blucher.
  4. Meirovitch, L., 2003. *Methods of Analytical Dynamics*. Dover Publications.
  5. Pesce, C.P., 1999. *Dinâmica dos corpos rígidos*.
- A complete set of lectures on Analytical Mechanics can be found here;
- These notes are accompanied from following Julia files: `JuliaBasics.ipynb` and `LinearModes.jl`

# Outline

# Fundamental concepts

- Mechanical system: System of interacting particles;
- Configuration space: Let $N$ be the number of independent particles (free of constraints) of a certain mechanical system. We can define a space (configuration space) characterized by the coordinates $x_i$, $y_i$ and $z_i$ ($i = 1, 2 \ldots, N$) of each particle. The temporal evolution of the mechanical system is a curve in the configuration space;
- Constraint equations: Commonly, we have constraint equations that link the motion of some particles. We define $c$ as the number of constraint equations;
- Focus is placed on holonomic constraints (constraint equations depend on the generalized coordinates and not on the generalized velocities). $f_k = f_k(x_i, y_i, z_i, t)$ $(k = 1, \ldots, c; i = 1, \ldots, N)$→ Holonomic and rheonomic constraint. $f_k = f_k(x_i, y_i, z_i)$ $(k = 1, \ldots, c; i = 1, \ldots, N)$→ Holonomic and scleronomic constraint and focus of the class.
- Number of degrees of freedom (dof): Number of generalized coordinates necessary for the complete description of the mechanical system. $n = 3N - c$;

# Fundamental concepts

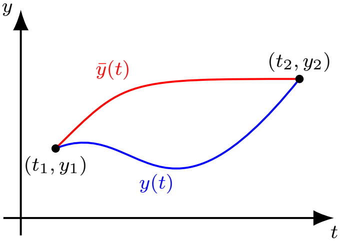
- Generalized coordinate $q_i$: Set of independent variables that defines the configuration of a mechanical system. The temporal derivative of the generalized coordinates corresponds to the generalized velocities. The generalized coordinates must define, in a biunivic way, the motion in the physical coordinates. The choice of the generalized coordinates is not unique in general.;
- Virtual displacement: Arbitrary change in the position of the particles that satisfies the constraints of the problem. The virtual displacement does not consider the flow of time. For a system with $n$ dofs described by $q_1, q_2 \ldots, q_n$, if the position of a certain particle is $\boldsymbol{r}_i = \boldsymbol{r}_i(q_1, q_2, \ldots, q_n, t)$, the associated virtual displacement is

$$\delta \boldsymbol{r}_i = \sum_{j=1}^{n} \frac{\partial \boldsymbol{r}_i}{\partial q_j} \delta q_j \tag{1}$$

where $\delta q_j$ is the variation of the generalized coordinate $q_j$.

# Basics of the calculus of variations

- For the purpose of this course, we can define a functional as a map from the space of functions to the real numbers;
- Consider the function $y = y(t)$ and the functional $\mathcal{F}[y, \dot{y}, t]$. In addition, consider modified functions $\bar{y}(t) = y(t) + \epsilon\eta(t) = y(t) + \delta y$.
- In this definition, it is assumed that $\bar{y}(t_1) = y(t_1) = y_1$ and $\bar{y}(t_2) = y(t_2) = y_2$ (see figure at right). Hence, $\eta(t_1) = \eta(t_2) = 0$ and $\delta y(t_1) = \delta y(t_2) = 0$
- Notice that $\dot{\bar{y}}(t) = \dot{y}(t) + \epsilon\dot{\eta}(t) = \dot{y}(t) + \delta\dot{y}$;
- In the above definition, $\delta y$ and $\delta \dot{y}$ are the variation of $y(t)$ and $\dot{y}$, respectively;
- Let's see what happens to the functional when we consider a change in the functions $y(t)$ and $\dot{y}(t)$.



# Basics of the calculus of variations

- Using Taylor series and keeping just the term linear in $\epsilon$

$$\mathcal{F}[y+\delta y,\dot{y}+\delta\dot{y},t]-\mathcal{F}[y,\dot{y},t]=\delta\mathcal{F}=\frac{\partial\mathcal{F}}{\partial y}\delta y+\frac{\partial\mathcal{F}}{\partial\dot{y}}\delta\dot{y} \tag{2}$$

- $\delta\mathcal{F}$ is the variation of $\mathcal{F}[y,\dot{y},t]$. Similarly to the differential calculus, a functional is said to be stationary at $y$ if $\delta\mathcal{F}=0$ (the functional does not change its signal in the neighborhood of $y$);
- A problem of importance is to compute the stationary value of a definite integral. For this, consider the functional $\mathcal{H}=\int_{t_1}^{t_2}\mathcal{F}[y,\dot{y},t]dt$:

$$\begin{aligned}\delta\mathcal{H}&=\int_{t_1}^{t_2}\mathcal{F}[y+\delta y,\dot{y}+\delta\dot{y},t]dt-\int_{t_1}^{t_2}\mathcal{F}[y,\dot{y},t]dt=\\&=\int_{t_1}^{t_2}(\mathcal{F}[y+\delta y,\dot{y}+\delta\dot{y},t]-\mathcal{F}[y,\dot{y},t])dt=\int_{t_1}^{t_2}\left(\frac{\partial\mathcal{F}}{\partial y}\delta y+\frac{\partial\mathcal{F}}{\partial\dot{y}}\delta\dot{y}\right)dt\end{aligned} \tag{3}$$

- In the above calculations, just the terms that are linear in $\epsilon$ are kept in the Taylor expansion. For removing the term $\delta\dot{y}$ in Eq. (3), we use the integration by parts:

$$\int_{t_1}^{t_2}\left(\frac{\partial\mathcal{F}}{\partial y}\delta y+\frac{\partial\mathcal{F}}{\partial\dot{y}}\delta\dot{y}\right)dt=\frac{\partial\mathcal{F}}{\partial\dot{y}}\delta y|_{t_1}^{t_2}-\int_{t_1}^{t_2}\left(\frac{\partial\mathcal{F}}{\partial y}\delta y-\frac{d}{dt}\left(\frac{\partial\mathcal{F}}{\partial\dot{y}}\right)\delta y\right)dt \tag{4}$$

# Basics of the calculus of variations

- Recalling that $\delta y(t_1)=\delta y(t_2)=0$ and that the stationary value of a functional implies its variation as null, the function $y=y(t)$ must satisfy:

$$\int_{t_1}^{t_2}\left(\frac{\partial \mathcal{F}}{\partial y}-\frac{d}{dt}\left(\frac{\partial \mathcal{F}}{\partial \dot{y}}\right)\right)\delta y dt=0 \tag{5}$$

- Since $\delta y$ is an arbitrary function with $\delta y(t_1)=\delta y(t_2)=0$, the above integral holds if:

$$\frac{\partial \mathcal{F}}{\partial y}-\frac{d}{dt}\left(\frac{\partial \mathcal{F}}{\partial \dot{y}}\right)=0 \tag{6}$$

- Equation (6) is known as Euler-Lagrange's equation and can be employed for obtaining the function $y(t)$ corresponding to the stationary value of the functional $\mathcal{H}$.

# Basics of the calculus of variations

- Now, we see an alternative way to compute the first variation of a functional. The present description is based on the lectures taught by Prof. de Zagottis.
- We consider the functional $\mathcal{F}[y, \dot{y}, t]$. We define $f(\alpha)$ as a real-valued function given by $f(\alpha) = \mathcal{F}[y + \alpha\delta y, \dot{y} + \alpha\delta\dot{y}, t]$.
- Expanding $f(\alpha)$:

$$f(\alpha) = f(0) + \left(\frac{df}{d\alpha}\right)_{\alpha=0} \alpha + \frac{1}{2!}\left(\frac{d^2 f}{d\alpha^2}\right)_{\alpha=0} \alpha^2 + \frac{1}{3!}\left(\frac{d^3 f}{d\alpha^3}\right)_{\alpha=0} \alpha^3 + \dots \tag{7}$$

- Recalling that $f(1) - f(0) = \mathcal{F}[y + \delta y, \dot{y} + \delta\dot{y}, t] - \mathcal{F}[y, \dot{y}, t]$, Eq. (7) is evaluated at $\alpha = 1$ leading to:

$$\mathcal{F}[y + \delta y, \dot{y} + \delta\dot{y}, t] - \mathcal{F}[y, \dot{y}, t] = \left(\frac{df}{d\alpha}\right)_{\alpha=0} + \frac{1}{2!}\left(\frac{d^2 f}{d\alpha^2}\right)_{\alpha=0} + \frac{1}{3!}\left(\frac{d^3 f}{d\alpha^3}\right)_{\alpha=0} + \dots =$$
$$= \delta\mathcal{F} + \frac{1}{2!}\delta^2\mathcal{F} + \frac{1}{3!}\delta^3\mathcal{F} + \dots \tag{8}$$

# Basics of the calculus of variations

- Consider the functional $\mathcal{F}[y, \dot{y}, t] = \frac{1}{2}m\dot{y}^2 - \frac{1}{2}ky^2$;
- It is easy to observe that:

$$f(\alpha) = \frac{1}{2}m(\dot{y} + \alpha\delta\dot{y})^2 - \frac{1}{2}k(y + \alpha\delta y)^2 \tag{9}$$

and

$$\frac{df}{d\alpha} = \frac{1}{2}m(2\dot{y}\delta\dot{y} + 2\alpha(\delta\dot{y})^2) - \frac{1}{2}k(2y\delta y + 2\alpha(\delta y)^2) \tag{10}$$

- Using Eq. (8), we obtain $\delta\mathcal{F} = \left(\frac{df}{d\alpha}\right)_{\alpha=0} = m\dot{y}\delta\dot{y} - ky\delta y$.
- The above result can also be obtained using Eq. (2).

# d'Alembert's principle

- Consider a system with $N$ point masses $m_i, i = 1, 2, \ldots, N$ defined by the corresponding position vectors $\boldsymbol{r_i}$.
- Second Newton's law, assuming that $m_i$ is independent of time:

$$\frac{d}{dt}(m_i\dot{\boldsymbol{r}}_i) = \boldsymbol{F}_i \rightarrow m_i\ddot{\boldsymbol{r}}_i = \boldsymbol{F}_i = \boldsymbol{F}_{i,a} + \boldsymbol{F}_{i,ic} + \boldsymbol{F}_{i,ni} \tag{11}$$

$\boldsymbol{F}_{i,a}$ is the applied force, $\boldsymbol{F}_{i,ic}$ and $\boldsymbol{F}_{i,ni}$ are the forces associated with ideal and non-ideal constraints.

- Restricted d'Alembert's principle:

$$-m_i\ddot{\boldsymbol{r}}_i + \boldsymbol{F}_{i,a} + \boldsymbol{F}_{i,ic} + \boldsymbol{F}_{i,ni} = \boldsymbol{0} \tag{12}$$

- $(-m_i\ddot{\boldsymbol{r}}_i + \boldsymbol{F}_{i,a} + \boldsymbol{F}_{i,ic} + \boldsymbol{F}_{i,ni}) \cdot \delta\boldsymbol{r}_i = 0$
- As the virtual work of the forces associated with ideal constraints is null and defining the effective force $\boldsymbol{F}_{i,e} = \boldsymbol{F}_{i,a} + \boldsymbol{F}_{i,ni}$, we have $(-m_i\ddot{\boldsymbol{r}}_i + \boldsymbol{F}_{i,e}) \cdot \boldsymbol{\delta r}_i = 0$
- For the system with $N$ particles

$$\sum_{i=1}^{N}(-m_i\ddot{\boldsymbol{r}}_i + \boldsymbol{F}_{i,e}) \cdot \delta\boldsymbol{r}_i = 0 \tag{13}$$

# Extended Hamilton's principle

- We define the effective force as the sum of a term arisen from a potential function (conservative force $\boldsymbol{F}_{i,c}$) with a non-conservative one $\boldsymbol{F}_{i,nc}$;
- The virtual work of the conservative and non-conservative forces are $\boldsymbol{F}_{i,c} \cdot \delta \boldsymbol{r}_i = -\delta \mathcal{U}_i$ and $\boldsymbol{F}_{i,nc} \cdot \boldsymbol{\delta r}_i = \delta W_{i,nc}$, respectively;
- With the above definitions, Eq. (13) reads

$$\sum_{i=1}^{N} (-\delta \mathcal{U}_i + \delta W_{i,nc} - m_i \ddot{\boldsymbol{r}}_i \cdot \delta \boldsymbol{r}_i) = 0 \rightarrow -\delta \mathcal{U} + \delta W_{nc} - \sum_{i=1}^{N} m_i \ddot{\boldsymbol{r}}_i \cdot \delta \boldsymbol{r}_i = 0 \tag{14}$$

- The variation of kinetic energy is

$$\delta \mathcal{T} = \sum_{i=1}^{N} \delta \left( \frac{1}{2} m_i \dot{\boldsymbol{r}}_i \cdot \dot{\boldsymbol{r}}_i \right) = \sum_{i=1}^{N} m_i \dot{\boldsymbol{r}}_i \cdot \delta \dot{\boldsymbol{r}}_i \tag{15}$$

- Mathematical identity:

$$\sum_{i=1}^{N} m_i \frac{d}{dt} (\dot{\boldsymbol{r}}_i \cdot \delta \boldsymbol{r}_i) = \sum_{i=1}^{N} (m_i \ddot{\boldsymbol{r}}_i \cdot \delta \boldsymbol{r}_i + m_i \dot{\boldsymbol{r}}_i \cdot \delta \dot{\boldsymbol{r}}_i) \tag{16}$$

# Extended Hamilton's principle

- From Eq. (16), we have

$$-\sum_{i=1}^{N} m_i \ddot{\boldsymbol{r}}_i \cdot \delta \boldsymbol{r}_i = -\sum_{i=1}^{N} m_i \frac{d}{dt}(\dot{\boldsymbol{r}}_i \cdot \delta \boldsymbol{r}_i) + \delta \mathcal{T} \tag{17}$$

- Using Eq. (17) in Eq. (14)

$$-\delta \mathcal{U} + \delta W_{nc} + \delta \mathcal{T} = \sum_{i=1}^{N} m_i \frac{d}{dt}(\dot{\boldsymbol{r}}_i \cdot \delta \boldsymbol{r}_i) \tag{18}$$

- Now, we integrate Eq. (18) from $t_1$ to $t_2$. In these instants, all the $\boldsymbol{r}_i$ are known and, hence $\delta \boldsymbol{r}_i(t_1)=\delta \boldsymbol{r}_i(t_2) = 0$

$$\int_{t_1}^{t_2} (\delta \mathcal{T} - \delta \mathcal{U} + \delta W_{nc}) dt = \sum_{i=1}^{N} m_i [\boldsymbol{r_i} \cdot \delta \boldsymbol{r_i}]_{t_1}^{t_2} = 0 \tag{19}$$

# Extended Hamilton's principle

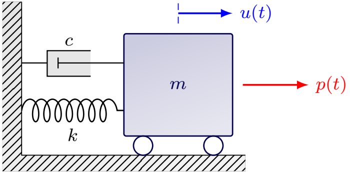
- Contrary to Newtonian mechanics, the use of analytical mechanics allows obtaining the equations of motion from the scalar quantities kinetic energy $\mathcal{T}$, potential energy $\mathcal{U}$ and virtual work of the non-conservative forces $\delta W_{nc}$;
- As shown (and discussed in greater depth in other graduate courses), the equations of motion are obtained by imposing

$$\int_{t_1}^{t_2} (\delta\mathcal{T} - \delta\mathcal{U} + \delta W_{nc})dt = 0 \tag{20}$$

where $\delta\mathcal{T}$ and $\delta\mathcal{U}$ are the first variations of kinetic and potential energies, respectively.
- The focus herein is not the derivation of the extended Hamilton's principle. The objective of this course is on the practical use of this principle for obtaining equations of motion of mechanical systems.

# Extended Hamilton's principle: application

We will find the equation of motion for the problem below sketched.




- Kinetic energy and its variation: $\mathcal{T} = \frac{1}{2}m\dot{u}^2 \rightarrow \delta\mathcal{T} = m\dot{u}\delta\dot{u}$;
- Potential energy and its variation: $\mathcal{U} = \frac{1}{2}ku^2 \rightarrow \delta\mathcal{U} = ku\delta u$;
- Virtual work of the non-conservative forces: $\delta W_{nc} = (-c\dot{u} + p(t))\delta u$

- Extended Hamilton's principle:

$$\int_{t_1}^{t_2} (\delta\mathcal{T} - \delta\mathcal{U} + \delta W^{nc})dt = 0 \leftrightarrow$$
$$\leftrightarrow \int_{t_1}^{t_2} [m\dot{u}\delta\dot{u} - (ku + c\dot{u} - p(t))\delta u]dt = 0 \quad (21)$$

- Integrating Eq. (21) by parts and recalling that $\delta u(t_1) = \delta u(t_2) = 0$, we obtain:

$$\int_{t_1}^{t_2} (-m\ddot{u} - c\dot{u} - ku + p(t))\delta u dt + \\ + \underbrace{[m\dot{u}\delta u]_{t_1}^{t_2}}_{0} = 0 \quad (22)$$

- Since $\delta u$ is arbitrary, Eq. (22) holds if $m\ddot{u} + c\dot{u} + ku = p(t)$.

# Euler-Lagrange's equation

- The kinetic energy $\mathcal{T}$ is function of the generalized coordinates $q_i$ and the corresponding generalized velocities $\dot{q}_i$. In some problems, it also explicitly depends on time. Mathematically, $\mathcal{T}$ and $\delta\mathcal{T}$ read

$$\mathcal{T} = \mathcal{T}(q_1, q_2, \ldots, q_n, \dot{q}_1, \dot{q}_2 \ldots, \dot{q}_n, t) \rightarrow \delta\mathcal{T} = \sum_{i=1}^{n} \left( \frac{\partial \mathcal{T}}{\partial q_i} \delta q_i + \frac{\partial \mathcal{T}}{\partial \dot{q}_i} \delta \dot{q}_i \right) \tag{23}$$

- On the other hand, the potential energy is function of the generalized coordinates.

$$\mathcal{U} = \mathcal{U}(q_1, q_2, \ldots, q_n) \rightarrow \delta\mathcal{U} = \sum_{i=1}^{n} \frac{\partial \mathcal{U}}{\partial q_i} \delta q_i \tag{24}$$

- Consider that the mechanical system is loaded by $N_f$ non-conservative forces $\boldsymbol{F_j}, j = 1, 2, \ldots, N_f$. The virtual work of the non-conservative forces is:

$$\delta W_{nc} = \sum_{j=1}^{N_f} \boldsymbol{F_j} \cdot \boldsymbol{\delta r_j} = \sum_{j=1}^{N_f} \boldsymbol{F_j} \cdot \sum_{i=1}^{n} \frac{\partial \boldsymbol{r_j}}{\partial q_i} \delta q_i = \sum_{i=1}^{n} \underbrace{\left( \sum_{j=1}^{N_f} \boldsymbol{F_j} \cdot \frac{\boldsymbol{\partial r_j}}{\partial q_i} \right)}_{Q_i} \delta q_i \tag{25}$$

# Euler-Lagrange's equation

- Equations (23)-(25) are substituted into the extended Hamilton's principle. Using integration by parts and recalling that $\delta q_i(t_1) = \delta q_i(t_2) = 0, i = 1, 2, \ldots, n$, we have:

$$\int_{t_1}^{t_2} \sum_{i=1}^{n} \left[ -\frac{d}{dt} \left( \frac{\partial \mathcal{T}}{\partial \dot{q}_i} \right) + \frac{\partial \mathcal{T}}{\partial q_i} - \frac{\partial \mathcal{U}}{\partial q_i} + Q_i \right] \delta q_i = 0 \quad (26)$$

- Equation (26) holds if:

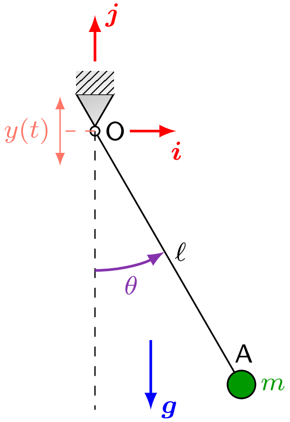
$$\frac{d}{dt} \left( \frac{\partial \mathcal{T}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{T}}{\partial q_i} + \frac{\partial \mathcal{U}}{\partial q_i} = Q_i, i = 1, 2, \ldots, n \quad (27)$$

- Equation (27) is known as Euler-Lagrange's equation. If we define the Lagrangian as $\mathcal{L} = \mathcal{T} - \mathcal{U}$, Eq. (27) is rewritten as:

$$\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i, i = 1, 2, \ldots, n \quad (28)$$

# Pendulum under support excitation

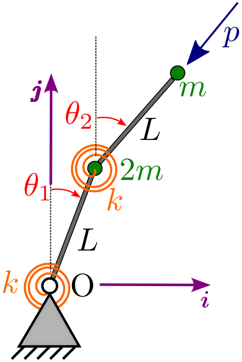
Obtain the equation of motion for the pendulum under support excitation. The rigid and massless arm has length $\ell$. Vertical displacement $y(t)$ is applied to the support




- Velocity of the mass:
  $\boldsymbol{v_m} = \boldsymbol{v_O} + \boldsymbol{\omega} \times (\boldsymbol{m} - \boldsymbol{O}) =$
  $\dot{y}\boldsymbol{j} + \dot{\theta}\boldsymbol{k} \times (\ell \sin\theta \boldsymbol{i} - \ell \cos\theta \boldsymbol{j}) = (\dot{\theta}\ell\cos\theta)\boldsymbol{i} + (\dot{y} + \dot{\theta}\ell\sin\theta)\boldsymbol{j}$;
- The origin of the fixed referential coincides with the hinge point O when $y(t) = 0$;
- Kinetic energy:
  $\mathcal{T} = \frac{1}{2} m \boldsymbol{v_m} \cdot \boldsymbol{v_m} = \frac{1}{2} m(\dot{y}^2 + 2\dot{y}\dot{\theta}\ell\sin\theta + (\dot{\theta}\ell)^2)$;
- Potential energy: $\mathcal{U} = mg(y - \ell\cos\theta)$;
- Virtual work of the non-conservative forces:
  $\delta W_{nc} = 0 \to Q_\theta = 0$
- Since $y(t)$ is prescribed, the system has one degree of freedom. In this case, we choose $\theta$ as the generalized coordinate;
- Using Euler-Lagrange's equation:
  $m\ell^2\ddot{\theta} + m(g + \ddot{y})\ell\sin\theta = 0$

# Ziegler's column under follower force

Classical problem, recently readdressed in d'Annibale & Ferretti (2020) and in Franzini & Mazzilli (2021). The massless bars have length $L$ and are connected by means of torsional springs of stiffness $k$. A follower force $p$ is applied to the tip.



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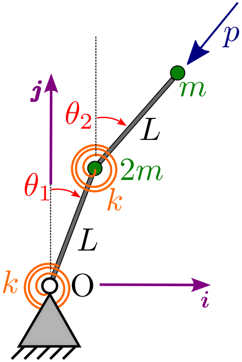



- Position vectors of the masses: $\boldsymbol{r}_1 = L(\sin\theta_1 \boldsymbol{i} + \cos\theta_1 \boldsymbol{j})$ and $\boldsymbol{r}_2 = \boldsymbol{r}_1 + L(\sin\theta_2 \boldsymbol{i} + \cos\theta_2 \boldsymbol{j})$;
- $\delta\boldsymbol{r}_1 = L(\cos\theta_1 \boldsymbol{i} - \sin\theta_1 \boldsymbol{j})\delta\theta_1$;
- $\delta\boldsymbol{r}_2 = \delta\boldsymbol{r}_1 + L(\cos\theta_2 \boldsymbol{i} - \sin\theta_2 \boldsymbol{j})\delta\theta_2 =$;
- Kinetic energy: $\mathcal{T} = \frac{1}{2}(2m)\dot{\boldsymbol{r}}_1 \cdot \dot{\boldsymbol{r}}_1 + \frac{1}{2}m\dot{\boldsymbol{r}}_2 \cdot \dot{\boldsymbol{r}}_2$
- Potential energy: $\mathcal{U} = \frac{1}{2}k\theta_1^2 + \frac{1}{2}k(\theta_2 - \theta_1)^2$;

# Ziegler's column under follower force

- Virtual work of the non-conservative force:

$$\begin{aligned}
&\delta W_{nc} = \boldsymbol{p} \cdot \delta \boldsymbol{r}_2 = \\
&= -p(\sin\theta_2 \boldsymbol{i} + \cos\theta_2 \boldsymbol{j}) \cdot L((\cos\theta_1 \delta\theta_1 + \cos\theta_2 \delta\theta_2)\boldsymbol{i} - (\sin\theta_1 \delta\theta_1 + \sin\theta_2 \delta\theta_2)\boldsymbol{j}) = \\
&= -pL\sin(\theta_2 - \theta_1)\delta\theta_1 \rightarrow Q_{\theta_1} = -pL\sin(\theta_2 - \theta_1), Q_{\theta_2} = 0 \qquad (29)
\end{aligned}$$

- After the derivatives and the application of Euler-Lagrange equation:

$$\begin{aligned}
&3mL^2\ddot{\theta}_1 + mL^2\cos(\theta_2 - \theta_1)\ddot{\theta}_2 + mL^2\sin(\theta_1 - \theta_2)\dot{\theta}_2^2 + 2k\theta_1 - k\theta_2 = \\
&= pL\sin(\theta_1 - \theta_2) \qquad (30) \\
&mL^2\cos(\theta_1 - \theta_2)\ddot{\theta}_1 + mL^2\ddot{\theta}_2 + k\theta_2 - k\theta_1 - mL^2\sin(\theta_1 - \theta_2)\dot{\theta}_1^2 = 0 \qquad (31)
\end{aligned}$$

# Ziegler's column under follower force

Now, we linearize the equations of motion around a certain position $(\theta_1^0; \theta_2^0)$. We consider $\theta_1(t) = \theta_1^0 + q_1(t)$ and $\theta_2(t) = \theta_2^0 + q_2(t)$, $q_1(t)$ and $q_2(t)$ disturbances superimposed to the point around which the mathematical model is linearized.

- $\dot{\theta}_1 = \dot{q}_1$, $\dot{\theta}_2 = \dot{q}_2$, $\ddot{\theta}_1 = \ddot{q}_1$ and $\ddot{\theta}_2 = \ddot{q}_2$;
- Using Taylor series and keeping only the first two terms: $\sin(\theta_1 - \theta_2) = \sin(\theta_1^0 - \theta_2^0) + \cos(\theta_1^0 - \theta_2^0)q_1 - \cos(\theta_1^0 - \theta_2^0)q_2$ and $\cos(\theta_1 - \theta_2) = \cos(\theta_1^0 - \theta_2^0) - \sin(\theta_1^0 - \theta_2^0)q_1 + \sin(\theta_1^0 - \theta_2^0)q_2$;
- Here, we linearize around the trivial condition $(\theta_1^0; \theta_2^0) = (0; 0)$, we have $\sin(\theta_1 - \theta_2) = q_1 - q_2$ and $\cos(\theta_1 - \theta_2) = 1$.
- Using these quantities, the linearized mathematical model is:

$$3mL^2\ddot{q}_1 + mL^2\ddot{q}_2 + (2k - pL)q_1 - (k - pL)q_2 = 0 \tag{32}$$

$$mL^2\ddot{q}_1 + mL^2\ddot{q}_2 - kq_1 + kq_2 = 0 \tag{33}$$

- In this problem, the linearized stiffness matrix is no longer symmetric.

# Outline

# 1-dof linear oscillator

- As mentioned, the equation of motion of a 1-dof linear oscillator can be obtained by using either the second Newton's law or concepts of analytical mechanics (extended Hamilton principle, Euler-Lagrange's equation);
- Consider the 1-dof system of mass $m$, spring stiffness $k$ and damping constant $c$. The system is forced by an external load $p(t)$. The displacement (generalized coordinate) is $u = u(t)$. The equation of motion is given by:

$$m\ddot{u} + c\dot{u} + ku = p(t) \tag{34}$$

- Equation (34) is a second-order, linear, ordinary differential equation. The solution of Eq. (34) is the sum of the homogeneous solution $u_h$ with the particular solution $u_p$. Mathematically, $u = u_h + u_p$. The initial conditions $u(0) = u_0$ and $\dot{u}(0) = \dot{u}_0$ must be imposed to the complete solution;
- Definitions $\omega = \sqrt{\frac{k}{m}}$ (undamped natural frequency), $\xi = \frac{c}{2m\omega} = \frac{c}{2\sqrt{km}}$ (damping ratio) and $\omega_d = \omega\sqrt{1-\xi^2}$ (damped natural frequency).

# 1-dof linear oscillator: Free vibrations

- In free vibrations, the external oscillation is null $\rightarrow p(t) = 0$. Oscillations occur due to non-trivial initial conditions;
- Three cases appear, depending on the value of $\xi$. $\xi < 1$ (sub-critical case) is the focus herein, since it is more common on engineering problems (civil, mechanical and naval engineering);
- In the sub-critical case: $u(t) = \rho e^{-\xi \omega t} \cos(\omega_d t - \theta)$, $\rho$ and $-\pi/2 \leq \theta \leq \pi/2$ depending on the initial conditions. Oscillations amplitudes exponentially decay with time.
- A system in free vibrations oscillates with its damped natural frequency $\omega_d$!!!!;
- If $\xi \ll 1 \rightarrow \omega_d \approx \omega$. For example, $\xi = 0.10 \rightarrow \omega_d = 0.995\omega$.

# 1-dof linear oscillator: Harmonically forced vibrations

- In this case $p(t) = p_0 \cos(\bar{\omega} t) = Re\{p_0 e^{\mathbb{i}\bar{\omega} t}\}$, with $\mathbb{i} = \sqrt{-1}$ being the imaginary unit;
- In the damped case, the homogeneous solution $u_h \to 0$ when $t \to \infty$ (steady-state response). In this case only the particular solution $u_p$ appears;
- Finding $u_p$. We assume $u = u_p = Re\{U e^{\mathbb{i}\bar{\omega} t}\} \to \dot{u} = Re\{\mathbb{i}\bar{\omega} U e^{\mathbb{i}\bar{\omega} t}\}$, $\ddot{u} = Re\{-\bar{\omega}^2 U e^{\mathbb{i}\bar{\omega} t}\}$. Now, we consider the complex quantities in the equations of motion, taking the real part at the end of the derivation:
- Using the above consideration

$$(-m\bar{\omega}^2 + k + \mathbb{i}\bar{\omega} c) U e^{\mathbb{i}\bar{\omega} t} = p_0 e^{\mathbb{i}\bar{\omega} t} \to U = U(\bar{\omega}) = \frac{p_0}{(-m\bar{\omega}^2 + k + \mathbb{i}\bar{\omega} c)} = p_0 H(\bar{\omega}) \quad (35)$$

- $H(\bar{\omega})$ is the frequency response function. Notice that $H(\bar{\omega})$ and $U$ are complex functions.

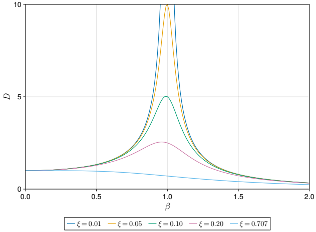
# 1-dof linear oscillator: Harmonically forced vibrations

Following, we define $\beta = \frac{\bar{\omega}}{\omega}$. As a consequence, $u$ reads:

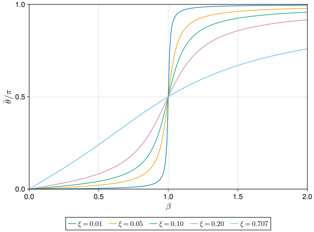
$$u = \frac{p_0}{(-m\bar{\omega}^2 + k + \mathbb{i}\bar{\omega}c)} e^{\mathbb{i}\bar{\omega}t} = \frac{p_0}{k}\left(\frac{1}{(1-\beta^2) + \mathbb{i}2\xi\beta}\right) e^{\mathbb{i}\bar{\omega}t} =$$
$$= \frac{p_0}{k}\left(\frac{1}{\sqrt{(1-\beta^2)^2 + (2\xi\beta)^2}}\right) e^{\mathbb{i}(\bar{\omega}t - \bar{\theta})}; \tan\bar{\theta} = \frac{2\xi\beta}{1-\beta^2} \tag{36}$$

- The dynamic magnification factor $D$ is defined as $D = \frac{1}{\sqrt{(1-\beta^2)^2+(2\xi\beta)^2}}$. Notice that $\frac{p_0}{k} = u_s$ is response to a load equal to the amplitude of the varying load statically applied to the structure. $0 \leq \bar{\theta} \leq \pi$ is the phase lag between excitation and displacement;
- As the external excitation is the real part of $p_0 e^{\mathbb{i}\bar{\omega}t}$, we take the real part of Eq. 36, obtaining $u = \bar{\rho}\cos(\bar{\omega}t - \bar{\theta}) = u_s D\cos(\bar{\omega}t - \bar{\theta})$.
- If the excitation is poly-chromatic (i.e., composed of a number of harmonic functions), we can obtain the response for each individual component and superimpose the result. This is valid within the linear theory!.

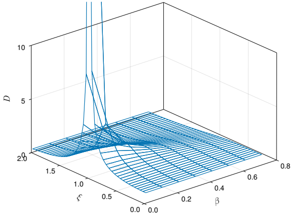
# 1-dof linear oscillator: Harmonically forced vibrations




(a) $D(\beta)$.

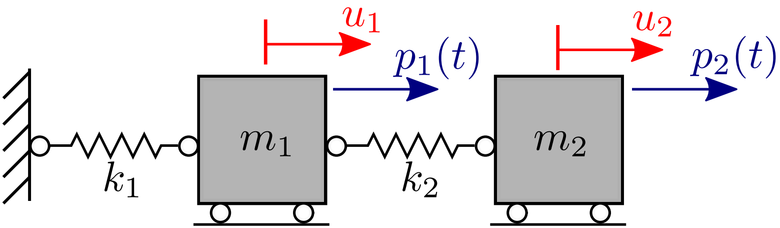



(b) $\bar{\theta}(\beta)$.




(c) $D(\xi, \beta)$.

# 2-dof undamped linear oscillator



# 2-dof undamped linear oscillator

In this case, the equations of motion are easily obtained by using Euler-Lagrange's equation.

- Kinetic energy:

$$\mathcal{T} = \frac{1}{2}m_1\dot{u}_1^2 + \frac{1}{2}m_2\dot{u}_2^2 \tag{37}$$

- Potential energy:

$$\mathcal{U} = \frac{1}{2}k_1u_1^2 + \frac{1}{2}k_2(u_2 - u_1)^2 \tag{38}$$

- Lagrangian $\mathcal{L} = \mathcal{T} - \mathcal{U}$;
- Virtual work of the non-conservative forces:

$$\delta W_{nc} = p_1\delta u_1 + p_2\delta u_2 = Q_1\delta u_1 + Q_2\delta u_2 \tag{39}$$

# 2-dof undamped linear oscillator

Euler-Lagrange's equation:

$$\frac{d}{dt}\left(\frac{d\mathcal{L}}{d\dot{u}_i}\right) - \frac{\partial \mathcal{L}}{\partial u_i} = Q_i, i = 1, 2 \tag{40}$$

$$m_1\ddot{u}_1 + (k_1 + k_2)u_1 - k_2 u_2 = p_1(t) \tag{41}$$

$$m_2\ddot{u}_2 - k_2 u_1 + k_2 u_2 = p_2(t) \tag{42}$$

We can define vectors $\boldsymbol{U} = \{u_1 \quad u_2\}^T$ and $\boldsymbol{P}(t) = \{p_1(t) \quad p_2(t)\}^T$. Using these definitions, Eqs. (41) and (42) can be written in the matrix form as

$$\boldsymbol{M}\ddot{\boldsymbol{U}} + \boldsymbol{K}\boldsymbol{U} = \boldsymbol{P}(t) \tag{43}$$

$\boldsymbol{M}$ and $\boldsymbol{K}$ being the mass matrix and the stiffness matrix, respectively. If linear damping is considered, the term $\boldsymbol{C}\dot{\boldsymbol{U}}$ is included on the left-hand side of Eq. (43).

# Modal analysis

- Any N-dof undamped linear oscillator is governed by the general Eq. (43). Modal analysis deals with the unforced response of this system ($\boldsymbol{P}(t) = \boldsymbol{0}$). In this scenario:

$$\ddot{\boldsymbol{U}} + \boldsymbol{A}\boldsymbol{U} = \boldsymbol{0} \tag{44}$$

  where $\boldsymbol{A} = \boldsymbol{M}^{-1}\boldsymbol{K}$.

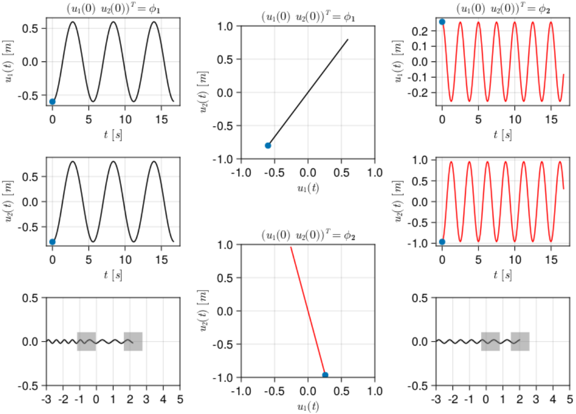
- A general solution for Eq. (44) is $\boldsymbol{U} = \boldsymbol{\phi} e^{\mathbb{i}\omega t}$. Substituting this expression into Eq. (44), we have:

$$(\boldsymbol{A} - \omega^2 \boldsymbol{I})\boldsymbol{\phi} e^{\mathbb{i}\omega t} = \boldsymbol{0} \tag{45}$$

- The existence of non-trivial solutions of Eq. (45) implies that $\det(\boldsymbol{A} - \omega^2 \boldsymbol{I}) = 0$. Hence, the eigenvalues of $\mathbf{A}$ are the natural frequencies $\omega$ squared. The eigenvectors $\boldsymbol{\phi}$ define the natural modes, associated with the "shape" of the response;
- The modal vectors $\boldsymbol{\phi}$ play a key role in dynamics. It can be shown that they can be used for decoupling a system of $N$ differential equations into $N$ uncoupled oscillators.

# Modal analysis - a numerical example

- $m_1 = 5$ kg, $m_2 = 1$ kg, $k_1 = 8$ N/m, $k_2 = 5$ N/m;
- $\lambda_1 = 1.2623 \Rightarrow \omega_1 = 1.1235$ rad/s, $T_1 = \frac{2\pi}{\omega_1} = 5.5924$ s;
- $\lambda_2 = 6.3377 \Rightarrow \omega_2 = 2.5175$ rad/s, $T_2 = \frac{2\pi}{\omega_2} = 2.4958$ s;
- Natural modes: $\phi_1 = \{0.5987\ 0.8009\}^T$ and $\phi_2 = \{0.2585\ \ -0.9660\}^T$;
- Modal matrix $\mathbf{\Phi} = [\phi_1\ \ \phi_2]$



# Response of a N-dof system under periodic load

- Equation of motion in matrix form:

$$\boldsymbol{M}\ddot{\boldsymbol{U}} + \boldsymbol{C}\dot{\boldsymbol{U}} + \boldsymbol{K}\boldsymbol{U} = \boldsymbol{P(t)} = \hat{\boldsymbol{P}}e^{\mathbb{i}\bar{\omega}t} \tag{46}$$

- We assume that the steady-state response is given by $\boldsymbol{U} = \hat{\boldsymbol{U}}e^{\mathbb{i}\bar{\omega}t}$. Using this assumption in Equation (46), one obtains:

$$\underbrace{\left(-\bar{\omega}^2\boldsymbol{M} + \boldsymbol{K} + \mathbb{i}\bar{\omega}\boldsymbol{C}\right)}_{\boldsymbol{Z}(\bar{\omega})}\hat{\boldsymbol{U}}e^{\mathbb{i}\bar{\omega}t} = \hat{\boldsymbol{P}}e^{\mathbb{i}\bar{\omega}t} \tag{47}$$

- $\boldsymbol{Z}(\bar{\omega})$ is the impedance matrix and $\boldsymbol{Z}^{-1}(\bar{\omega}) = \boldsymbol{H}(\bar{\omega})$ is the response frequency matrix. Notice that $\hat{\boldsymbol{U}} = \hat{\boldsymbol{U}}(\bar{\omega}) = \boldsymbol{H}(\bar{\omega})\hat{\boldsymbol{P}}$.
- The $k^{th}$ column of $\boldsymbol{H}(\bar{\omega})$ corresponds to the steady-state response amplitude of the different degrees of freedom to a harmonic excitation of unitary amplitude only on the $k^{th}$ degree of freedom.

# Outline

# State-space representation

- The representation of the dynamics on the configuration space does not allow assessing the velocities involved in the dynamics;
- Hence, the representation in the state-space appears as an interesting approach;
- As a first example, consider the forced 1-dof linear oscillator $\ddot{u} + 2\xi\omega\dot{u} + \omega^2 u = p(t)/m$. We define the state-variables as $x_1 = u$ and $x_2 = \dot{u}$.
- The original equation is rewritten as:

$$\dot{x}_1 = x_2 \tag{48}$$

$$\dot{x}_2 = -\omega^2 x_1 - 2\xi\omega x_2 + p(t)/m \tag{49}$$

- Defining $\boldsymbol{x} = \{x_1 \quad x_2\}^T$, Eqs. 48 and 49 can be rewritten in matrix form as:

$$\dot{\boldsymbol{x}} = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -2\xi\omega \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} + \begin{Bmatrix} 0 \\ p(t)/m \end{Bmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \tag{50}$$

- In a more general form $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{\mu}, t)$, $\boldsymbol{\mu} \in \mathbb{R}^p$ being a vector with parameters of the mathematical model (mass, damping, stiffness...). A system with this form is said to be non-autonomous, since it explicitly depends on time. An autonomous system has no explicit dependence on time and, hence, is given by the general form $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{\mu})$.

# State-space representation

- The state-space representation transforms an ordinary differential equation of second order into a system of first-order differential equations;
- For linear undamped systems with $N$ degrees of freedom, the general equation of motion is given by Eq. 43. If we define $\boldsymbol{x} = \{u_1 \quad u_2 \ldots u_N \quad \dot{u}_1 \quad \dot{u}_2 \ldots \dot{u}_N\}^T$, the corresponding first-order system of differential equations is:

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \mathbf{0}_{N\times N} & \boldsymbol{I}_{N\times N} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & \mathbf{0}_{N\times N} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{2N} \end{Bmatrix} + \begin{Bmatrix} \mathbf{0}_{N\times 1} \\ \boldsymbol{P}(t) \end{Bmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \tag{51}$$

- Notice that the damping matrix $\boldsymbol{C}$ can be easily included in Eq. (51).
- Notice also that in non-linear systems, $\boldsymbol{M}$, $\boldsymbol{C}$ or $\boldsymbol{K}$ depend on the state-vector $\boldsymbol{x}$;
- For non-autonomous systems, the dimension of the state-space is $2N$. On the other hand, this dimension is $2N+1$ for non-autonomous systems;
- It is worth mentioning that we can transform a non-autonomous system into an autonomous one. For this, we can include the state-variable $x_{N+1} = t$ and its derivative as $\dot{x}_{N+1} = 1$.

# Example: van der Pol equation

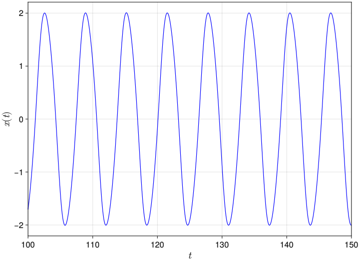
- Consider the van der Pol equation $\ddot{u} + \epsilon(u^2 - 1)\dot{u} + u = 0$
- Defining $x_1 = u$ and $x_2 = \dot{u}$, we have:

$$\dot{x}_1 = f_1(\boldsymbol{x}, \epsilon) = x_2 \quad (52)$$

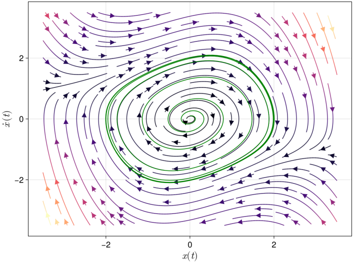
$$\dot{x}_2 = f_2(\boldsymbol{x}, \epsilon) = -x_1 - \epsilon(x_1^2 - 1)x_2 \quad (53)$$

- The steady-state solution of the van der Pol equation is a curve (orbit) in the phase-plane $(x_1(t); x_2(t))$. The tangent vector to this curve is $\{\dot{x}_1; \dot{x}_2\}^T = \{x_2; -x_1 - \epsilon(x_1^2 - 1)x_2\}^T$;
- The plot of the vector field $\boldsymbol{F} = \{f_1, f_2\}^T$ offers important insights into the organization of the phase space.

# Example: van der Pol equation




(a) $u(t) = x_1(t)$.




(b) $x_2(t) \times x_1(t)$.